

Deakin Breathes Data

Fabric(ating) Filaments of Knowledge and Campus Life

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Figure 1: Silk object made through the dialogue of digital and analogue processes, evidencing the entanglement that exists between the human hand work and non-human data signals in creative practices.

Abstract

Deakin Breathes Data (DBD) rethought and rethreaded radio surveillance on a university campus into a material, fibrous aesthetic. We used university-captured Wi-Fi data to present how shared knowledge threads the modern university experience. Laser cut silk mirrored how Wi-Fi's photonic wave-particles slice through spaces, mapping knowledge channels in and between buildings as they vary with student presence. Its contribution considers physical and metaphorical fibers, showing how a campus-as-organism breathes data, through student knowledge sharing—both consensual and surveilled. We hook the material fibers together in a knit of organizational surveillance, knowledge creation, and campus life, reimagined through tactility, fabric, and laser, while aesthetically referencing the invention of frequency hopping itself [17], which Wi-Fi depends on.

CCS Concepts

• **Human-centered computing** → **Information visualization**; **Empirical studies in visualization**; • **Applied computing** → **Media arts**.

Keywords

Surveillance, Wi-Fi, Data Physicalization, Design

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1 Context

Deakin University is a corporately organized entity with 3 physical campuses across Australia and one campus branded as being in 'the cloud'. Each of the four campuses relies on robust Wi-Fi signals to connect students and staff to each other, course material, and the wider internet. These connections offer knowledge, socialization, and are measured and monitored by University Services, a division of the non-teaching and non-research side of the University, branded in the Australian context as 'professional staff'. Our colleagues in the Digital Solutions team of University Services were able to share aggregate anonymized Wi-Fi connections through

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their own data privacy governance, insofar our process and outcomes could not be further computed. That is to say, the data's transfiguration from functional to aesthetic opened up opportunities for knowing how Deakin breathes its knowledge in material physicalizations [12] that might otherwise be veiled behind commercial trade secrets, or in more granular settings, student data privacy issues.

Previous attempts to systematize an aesthetic politics in data physicalization [3] seem to tack away from what Karl Popper might describe as 'clouds' and towards a normative call for 'clocks', regarding what 'researchers and designers should consider' [3], namely measuring: audience and production context, and somewhat ironically, tension between positivist/interpretivist representation. Here we see no need to measure context on one axis and epistemological/ontological awareness on another to determine the greatness of a data physicalization [25], but do recognise and reflect on how our own context influenced the work: ubiquitous, unseen data flows connect and produce campus life, affording novel formations of surveillance - knowledge - energy. The curation that DBD was imagined within highlighted ecosophy [19] or interconnected ecological harmony, allowing our choice of natural fiber (silk) to capture how the sharing and materialization of data is 'burned into' the natural world via fossil-fueled processes [23]. Moreover, automated laser cutting combining with manual hand threading of the work allowed us to reflect on the utility and aesthetic of Wi-Fi itself, as well as its context in larger educational, ecological, and politically economic surveillant designs as a double function artifact [9] that both captures and represents human and non-human intra-actions [4]. While knit data physicalization projects have focused on real time individualized interactions [9] [14] [13] or workshopped normative reorientations of technology [21] we chose a retrospective data signal of the university organization/organism – secondary data already gathered in surveillant campus design appropriated to new use.

The remainder of this paper first reviews the origin and uses and Wi-Fi from radiographic perspectives that shows our aesthetic lineage, and then offers how datafication ties it to current projects of university practice. We then offer the institutional context for our work before detailing the work process.

2 Wi-Fi Material Histories, Datafication, and Surveillant Use Designs

Here we introduce how Wi-Fi data came to be conceptualized and used, and how our material practice transfigures some of these assumptions to provide an aesthetic of critique and constructive tactility to experience modern universities. The history of what is commonly known as Wi-Fi shows how the underlying signal processing technology has transitioned purposes and been conceived in diverse ways. The radio frequency hopping that Wi-Fi is dependent on was invented as a "secret communication system" [17] to protect torpedoes and other guided munitions from German jamming for World War 2. This invention has a specific aesthetic quality that cut space into material to show where data could be transmitted (Figure 2).

Similar principles of radiography were later applied to increase image sharpness in medical applications [11] before the capacity of

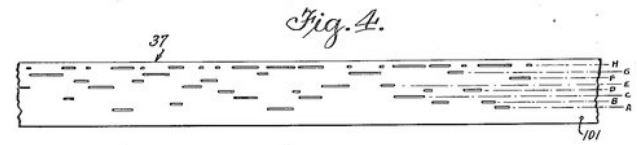


Figure 2: From “Markey HK, Anthel G. Secret communication system. US Patent 2,292,387. 1942 Aug 11.”

frequency hopping was applied to dramatically increase bandwidth of digital radio communication as “Wi-Fi” [20]. The implications of this latter widespread data transfer were social as much as technical. Wi-Fi functioned as new metaphorical threads of a networked society; imagined communities that broadcast at the national scale [1] could now scale up or down in decentralized ways, trading content and forming relations in real time where evolutions in hardware, software, and use cases created publics and more intimate groups for personal-social interaction [6] via wireless connection to the internet.

Our project considers aesthetics of such radio signals threading together our social existence. There have been numerous visualizations of the radio waves for planning purposes [22], while more artistic [15] or consumer applications show how radio and humanity is bound together, including novel applications in mobile-based Augmented Reality [24] that use Wi-Fi-radio enabled devices to surveil and visually and audibly render Wi-Fi and other radio sources in real time. However, our aesthetic goal was to move beyond common algorithms that spatialize - usually via heat map - signal reach or density, and to consider what Wi-Fi makes in specific contexts. We turned to the university setting to consider how Wi-Fi's surveillant, knowledge, community, and resource signals might be reconsidered materially and conceptually.

The institution's University Services digital division provided an avenue to do so, from a functionality metric they developed to count and track the amount of Wi-Fi devices connected in each building per campus, per day. These metrics filled a data need to surveil building use patterns modern institutions use as feedback loops across a spectrum of control to care [16] for decision making. On the one hand, these data show the relative importance of each building (and campus) by communicating student use of facilities. This is an important post-pandemic metric for Australian universities [7] who are dealing with stagnant funding, reduced international student income, online teaching normalization, and higher education restructuring [8] [18]. From this context, the Wi-Fi data was conceptualized functionally as space utilization, and shows the normalization of surveillance in 'student experience'; students commonly provide data points that are computed in Learning Management Systems (LMS) to measure and predict course completion, student evaluations around content and teaching, course and staffing demand planning; among other data feeds that show the automated surveillance of students and staff in higher education [10]. Concern towards how this surveillance can define 'innovation' tied to surveillance capitalism in higher education presents in learning platforms [2] to library analytics [5].

Surveillance is presented as instrumental in the above cases of higher education, as they point to logics of care and control for the

student body, the institution, or learning more abstractly. As such we reconsider what the aesthetic of such surveillance might look like, or be conceptualized as, when we deregister it from learning outcomes (broadly defined) and facilities use.

For instance, past surveillance, our aesthetic considerations include how data is actually *oil* [23], with data usage providing a direct link to the energy needs of learning in the digital age - not as a metaphor but as extractive material practice, measurable by hardware build outs and radio wave strength. The aesthetic beauty of this extraction is here burnt in silk, focussed through handwork, and displays undulating channels of community presence-knowledge that flow from one side of our material work to the other representing 365 days of the university organism. We chose to work with 100% silk to emphasise a natural fibre as way to bring into dialogue the environmental impact of Wi-Fi. On another note, the threads of shared knowledge are presented as absence - the ephemeral knowledge of shared space and classes burnt into the silk of the surrounding ecosphere and threading up through time, with hand-knitted material practice connecting each day to another, transforming an act of measuring into a process of aesthetic. The processes to accomplish these aesthetic goals are clarified below

3 Processes of Wi-Fi Materiality

3.1 The Data Process

The retrieved Wi-Fi connectivity data is in .csv format. It contains two different data sets: visits by campus and visits by building, separated by year according to when the data was collected. As their name suggests, the first data set contains a list of numbers displaying the amount of connected IP addresses per day per campus, while the second data set contains a list of numbers displaying the number of connected IP addresses per day per building on each campus. For this work, we selected the file containing the visits by building during the year 2022 at Deakin University's Burwood Campus.

As a first instance, we divided the data set into lists, each corresponding to a building. This process resulted in 73 sub-lists, each containing the corresponding connectivity data per day during the year 2022. However, we encountered that several days of data were missing from the lists.

In practical terms this meant that to ensure data integrity and computable data files, we developed a script that filled the blank spaces with the corresponding missing date and a number '0' to indicate zero connections that day and space. After running this script, we obtained 73 lists of data, one for each building, each containing 365 rows, one for each day of the year 2022, and filled with numbers ranging from zero to 5761, representing the minimum and maximum occupancy per building per day.

3.2 The Work Process

The authors decided early on in the process that the designed piece would include a media transformation from digital data to an analogue outcome. This approach, the materialisation of the digital, allows for particular and innovative aesthetics to appear, triggering the creation of new creative pipelines that influence and expand existing ones in design and making. The explorative process of using a list of numbers for the creation of analogue objects informs

both the designers/makers and the audience/viewers. Designers and makers are challenged by the redefinition and renegotiation of their established methodologies in their practices, while the audience/viewers engage in different ways of experiencing information, where the goal is not to understand the statistical value of the data, but through an artistic experience immerse themselves in other multilayered aspects that this data also represents.

In this work, the data defines the aesthetic outcome, and the designers/makers intervene with their handwork in the process of making with the digital and analogue materials. The data not only defines the shape but also proposes a framework for the possible pipelines that could make sense of it. Here, there is a hierarchical balance and interdependence between data, digital and analogue interfaces, and handwork that defines the possible aesthetics and usability of the designed object.

3.3 The Pipeline Process

For the design of the piece, we used the program Rhinoceros 3D and its parametric design plugin Grasshopper. This software allows for the creation of digital geometries that can be linked to data sets to manipulate their overall properties in real-time. The benefits of using this 3D modelling software are the possibility of using one script that can be reused for different data sets, the rendering tool offers a preview of the final piece without the need for physical prototyping, and the output file can be used for 3d printing, laser cut, CNC milling or other alternatives for the making of objects. This allows for a higher degree of freedom in creative and explorative practices of making with the digital.

After several exploration stages and small prototyping sampling, we decided to create a rectangular shape as a core unit to work with, and we chose laser cutting as an interface for the making. This geometrical shape allows for a reduced waste impact, maximizing the use of material. We chose to work with 100% silk as a material; choosing a natural fiber reduces the environmental impact of the work and creates a dialogue with the nature of the retrieved data. From people connecting on campus to digital numbers to natural fiber through machine and handwork processes.

Working in Grasshopper, we defined the rectangular shape with a size of 520mm x 30mm. These proportions were considered according to the size of the laser cutter's working area and the width of the selected material. Here, the material and interfaces have a direct impact and influence on the characteristics of the work, imprinting technological traces during the process that define the overall aesthetics of the piece.

The core unit contains the data of all the buildings on a specific date. Hence, 365 rectangles are needed to use all the data retrieved during 2022. Each rectangle is then subdivided into 73 sections according to each one of the buildings from the Burwood campus. The connectivity data is reparametrized on each building, where the maximum occupancy per building is considered 100% occupancy. Then, another rectangle or indentation is created on each building section, where the width of this indentation corresponds to the occupancy percentage. In this way, we created a strip containing the occupancy data per day per building (Figure 3). Using Grasshopper, we can repeat the process parametrically with the whole data set to distribute it on the 365 rectangles.

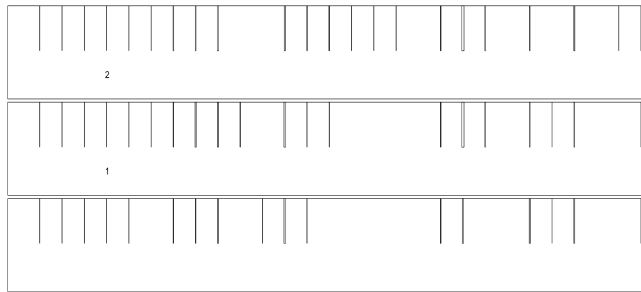


Figure 3: Rhinoceros 3D screenshot showing three rectangles with the correspondent indentations according to occupancy data per building, note similarities to Figure 1.

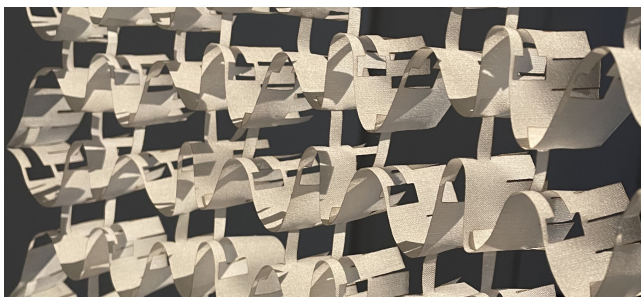


Figure 4: Cuts on the silk that allows for interlinking of the stripes without the need of other material, glue or sewing

3.4 The Building Process

Furthermore, and following a sustainable approach in the making, we created a set of cuts that work as hook and eye to link all the rectangles together without the need of any other material, glue, or sewing (Figure 4). This stage of the process required a delicate and precise handling of the material and a skilful dedication to connect the 3640 hooks from one strip to another. The linking of the silk stripes mirrors the hybrid characteristics of the making process, where the technology and the embodied knowledge of the designer/maker intertwine while making. The data underlying surveillance is transformed into an object to observe, grasp, and reflect.

The resulting piece is a 100% silk object that was made through the dialogue of digital and analogue processes, evidencing the entanglement that exists between the human hand work and non-human data signals in creative practices (Figure 1, 5)

4 Conclusion

This work rethreaded radiography on a university campus into (im)material fibres, linking surveillance, community, and knowledge, through a physical interplay of data and space through silk and the photonic. By transforming Wi-Fi data into tangible materials, we showcased how a campus-as-organism breathes data in ephemeral flows that reflect student use of space and networks of knowledge sharing. The material fibres of our work reveal a knit organization of surveillance, knowledge creation, and environmental impact, illustrating the flickering life of campus experience.



Figure 5: Final silk object

Our material aesthetic practice reimagines functional presentations of Wi-Fi data, providing a critique and constructive tactility to explain student experience as part of modern university, constructing intra-actions of space, energy, and knowledge. The project invites viewers to witness the university campus as vibrant threads, situating themes of ubiquitous surveillance - imagined through care, community, and control - behind the everyday practice of being connected. 'Deakin Breathes Data' is an invitation to imagine, reflect, and engage with our geo-digital traces, demonstrating how the materialization of data can transform our understanding of surveillance and connectivity in educational settings.

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